

Determine how many times faster the speed of light is than the speed of sound

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How Many Times Faster is the Speed of Light Than the Speed of Sound?

Key Points

- The speed of light in a vacuum is approximately (299792458) m/s, a well - known scientific constant.
- The speed of sound varies widely depending on the medium, being lowest in gases, higher in liquids, and highest in solids.
- In gases like air at 20C, the speed of light is about 871489.70 times faster than the speed of sound.
- In liquids such as fresh water at 20C, the speed of light is around 202562.47 times faster.
- In solids like aluminum, the speed of light is approximately 58553.21 times faster.

Overview

The comparison between the speed of light and the speed of sound is a fascinating topic in physics. The speed of light is an unchanging constant in a vacuum, while the speed of sound is highly variable, depending on the medium through which it travels. Understanding the difference in their speeds can help us grasp various natural phenomena, such as why we see lightning before we hear thunder. This research breaks down the comparison by different states of matter to provide a comprehensive view.

Detailed Analysis

Defining the Speed of Light

The speed of light in a vacuum is a fundamental scientific constant, approximately (299792458) m/s. This value serves as the basis for our comparison with the speed of sound in different media.

Speed of Sound in Different Media

The speed of sound varies significantly based on the medium. Here is a detailed look at the speed of sound in gases, liquids, and solids:

- **Gases**:
 - At 0C, the speed of sound in air is 331 m/s. For carbon dioxide, it is 259 m/s, in oxygen it is 316 m/s, in helium it is 965 m/s, and in hydrogen it is 1290 m/s.
 - At 20C, the speed of sound in air is 344 m/s. At normal conditions (15C and 1 atm), the speed of sound in air is approximately 340 m/s.
 - Factors like temperature and the type of gas affect the speed of sound in this state of matter.
 - !The content explains the speed of sound as the distance a sound wave travels through different states of matter, providing specific speeds at various temperatures and listing factors that affect sound speed.](https://sciencenotes.org/wp-content/uploads/2023/06/Speed-of-Sound-1024x683.png)
- **Liquids**:
 - At 20C, the speed of sound in ethanol is 1160 m/s, in mercury it is 1450 m/s, in fresh water it is 1480 m/s, in sea - water it is 1540 m/s, and in human tissue it is 1540 m/s.
 - The density and composition of the liquid play a role in determining the speed of sound.
 - !A table displaying the speed of sound in various media, including air, water, and iron, along with their respective velocities measured in meters per second.](http://stickmanphysics.com/wp-content/uploads/2020/03/Speed-of-Sound-in-Variou-Medium-5.jpg)
- **Solids**:
 - For vulcanized rubber, the speed of sound is 54 m/s, in polyethylene it is 920 m/s, in marble it is 3810 m/s, in Pyrex glass it is 5640 m/s, in lead it is 1960 m/s, and in aluminum it is 5120 m/s.
 - The molecular structure and rigidity of the solid influence the speed of sound.
 - !A table lists various media and their corresponding speeds, showing that sound travels at 343 m/s in air (20C, 1 atm), 1480 m/s in liquid water, and reaches speeds of 5120 m/s in solid iron.](https://echophysics.swarthmore.edu/wp-content/uploads/2020/07/Table1_soundspeeds.png)

Calculating the Ratios

To determine how many times faster the speed of light is than the speed of sound, we calculate the ratio of the speed of light to the speed of sound for each medium. Here are the results:

Medium	Ratio (Speed of Light / Speed of Sound)
air_0°C	905717.40
air_20°C	871489.70
air_15°C_1atm	881742.52
carbon_dioxide	1157499.84
oxygen	948710.31
helium	310665.76
hydrogen	232397.25
ethanol_20°C	258441.77
mercury_20°C	206753.42
fresh_water_20°C	202562.47

sea_water_20°C	194670.43
human_tissue_20°C	194670.43
vulcanized_rubber	5551712.19
polyethylene	325861.37
marble	78685.68
Pyrex_glass	53154.69
lead	152955.34
aluminum	58553.21

Survey Note

Thread Highlights

- "The speed of light in a vacuum is a constant, but the speed of sound varies wildly depending on the medium! "
- "In vulcanized rubber, the speed of light is over 5 million times faster than the speed of sound. Mind - blown! "

Data That Matters

- The speed of light in a vacuum: (299792458) m/s
- The ratio of the speed of light to the speed of sound in air at 20C: 871489.70

Community Pulse

This comparison between the speed of light and the speed of sound is a hot topic in the science community. Many are amazed by the vast differences in the ratios, especially in solids like vulcanized rubber.

Action Steps

- Share this thread with your science - loving friends to spread the knowledge.
- Dive deeper into the physics behind the speed of sound and light by reading the references provided.

Key Citations

- [17.3: Speed of Sound - Physics LibreTexts]([https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_\(OpenStax\)/Book%3A_University_Physics_I_-_Mechanics_Sound_Oscillations_and_Waves_\(OpenStax\)/17%3A_Sound/17.03%3A_Speed_of_Sound](https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_(OpenStax)/Book%3A_University_Physics_I_-_Mechanics_Sound_Oscillations_and_Waves_(OpenStax)/17%3A_Sound/17.03%3A_Speed_of_Sound))
- [Speed of sound - Oxford Reference](<https://www.oxfordreference.com/display/10.1093/oi/authority.20110803100522606>)

- [(PDF) Speed of Sound in Atmosphere of the Earth - ResearchGate](https://www.researchgate.net/publication/274750587_Speed_of_Sound_in_Atmosphere_of_the_Earth)

What do you think about these differences in the speed of light and sound? Retweet if you find this information fascinating!